

Heart Disease Prediction using Supervised Machine Learning: A Comparative Study Across Four Hospitals

CPIS-363 Intelligent Systems - Final Project (Spring 2026)

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Abstract

Cardiovascular disease remains the leading cause of death worldwide, and early automated screening can support clinical decision-making. In this project we build a complete supervised-learning pipeline on the UCI Heart Disease dataset, which combines records from four hospitals (Cleveland, Hungary, Switzerland, and VA Long Beach) and contains 918 unique patients after de-duplication. We address substantial missingness (up to 66% in some columns) using median and mode imputation, one-hot encode categorical variables, and apply stratified 80/20 train/test splitting with feature scaling fitted on the training set only to prevent data leakage. We train and tune four models - Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine, and a Random Forest ensemble - using 5-fold stratified cross-validation and grid-search hyperparameter optimization. On the held-out test set, KNN achieved the strongest balance with an accuracy of 87.5%, F1-score of 0.891, and ROC-AUC of 0.926, narrowly outperforming the Random Forest ensemble. A fairness analysis revealed performance gaps across both sex and source hospital, which we discuss as concrete ethical limitations of medical AI.

1. Introduction

Heart disease accounts for nearly one third of global mortality, and many cases remain undiagnosed until a serious cardiac event occurs. Clinical risk assessment relies on a combination of demographic information, symptoms, and non-invasive test results - exactly the kind of structured tabular data on which supervised machine-learning models excel. An accurate, transparent screening model could help triage patients in resource-limited settings and prioritize follow-up imaging or stress tests.

We frame the task as binary classification. The input to each algorithm is a feature vector of 13 clinical attributes per patient (e.g., age, sex, chest-pain type, resting blood pressure, cholesterol, maximum heart rate, ST depression, number of major vessels). The output is a single binary label: 1 if the patient has any degree of coronary disease, 0 otherwise. We convert the original UCI severity score num (0-4) into this binary target by mapping any non-zero severity to the positive class.

2. Dataset Description

We use the UCI Heart Disease dataset (Detrano et al., 1989) in its combined four-hospital form (920 raw rows, 918 after removing 2 exact duplicates). The dataset aggregates records from Cleveland Clinic Foundation (304), Hungarian Institute of Cardiology (292), University Hospital Zurich, Switzerland (123), and VA Medical Center Long Beach (199). The binary target distribution is mildly imbalanced: 508 patients with disease (55.3%) and 410 without (44.7%), giving a minority/majority ratio of 0.81 - close enough to balanced that stratified sampling alone is sufficient without resampling techniques like SMOTE.

The dataset is dominated by male patients ($725/918 = 79.0\%$), and missingness varies dramatically across features and hospitals. The Cleveland subset is nearly complete ($<1\%$ missing), while the VA Long Beach subset has roughly 20% missing values on average. Two features in particular have heavy missingness: ca (number of fluoroscopy-colored major vessels, 66.3% missing) and thal (thalassemia, 52.7% missing). These two columns are essentially unavailable for the Hungarian, Swiss, and VA cohorts because the relevant imaging tests were not performed.

Preprocessing was performed in the following order, all justified explicitly:

- Convert multi-class severity num (0-4) to binary target via (num > 0).
- Drop the patient id (no predictive value) and retain dataset for fairness analysis only.
- Drop exact duplicates (2 rows).
- Impute missing numeric values with the column median (robust to outliers) and missing categorical values with the column mode.
- One-hot encode the six categorical features (sex, dataset, cp, restecg, slope, thal) with drop_first=True to avoid the dummy-variable trap; final feature space has 21 columns.
- Stratified 80/20 train/test split (734 train, 184 test) preserving the class ratio.
- Standardize features using StandardScaler fitted on the training set only, then applied to the test set - preventing data leakage.

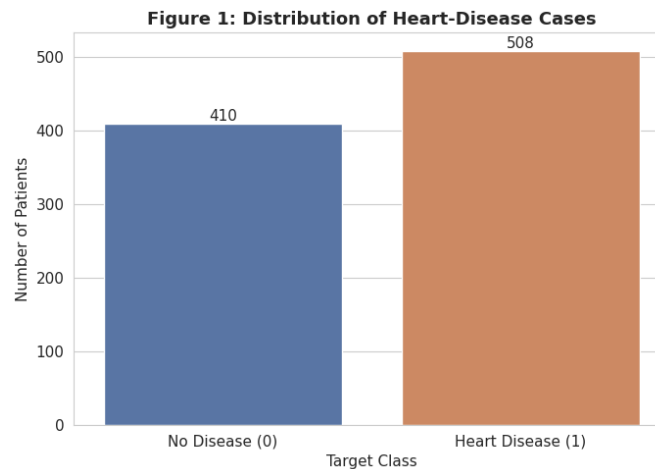


Figure 1. Target distribution after binarization: 508 patients with disease (55.4%) vs. 410 without (44.6%), a mild class imbalance handled by stratified splitting.

3. Methodology

We compare four supervised classifiers that span four distinct algorithmic families: linear, instance-based, margin-based, and tree-ensemble. This breadth lets us study how model assumptions interact with the cardiac data.

Logistic Regression.

A linear probabilistic model that estimates $P(\text{disease} \mid \mathbf{x})$ via the logistic (sigmoid) function applied to a weighted sum of features. Coefficients are learned by maximum likelihood with L2 regularization (controlled by C). The model assumes a roughly linear log-odds relationship between features and outcome and provides directly interpretable coefficients - valuable in clinical contexts.

K-Nearest Neighbors (KNN).

A non-parametric, instance-based classifier that assigns each test patient the majority class among its k closest training patients in standardized feature space. We tuned the neighborhood size (k), the weighting scheme (uniform vs. distance-weighted), and the Minkowski power ($p = 1$ for Manhattan, $p = 2$ for Euclidean). KNN makes no global assumption about the decision boundary.

Support Vector Machine (SVM).

An SVM seeks the maximum-margin hyperplane separating the two classes. With a non-linear kernel (we tried linear and RBF), it can model curved boundaries by implicitly mapping features into a higher-dimensional space. The C parameter trades off margin width vs. training-error tolerance. SVMs are strong on small-to-medium clean datasets but provide limited interpretability.

Random Forest (Ensemble).

Our ensemble method. A Random Forest trains many decision trees on bootstrap samples of the training data, each tree considering only a random subset of features at every split. Predictions are averaged across trees, which reduces variance dramatically while preserving the bias of individual trees. As a bonus, the forest yields a measure of feature importance, which we use to identify the strongest clinical predictors.

All four models were tuned with GridSearchCV using 5-fold stratified cross-validation on the training set, optimizing ROC-AUC. Final evaluation was performed once on the untouched test set with six metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC, and the Confusion Matrix.

Table 1. Best hyperparameters selected by 5-fold cross-validation.

Model	Best Hyperparameters	CV ROC-AUC
Logistic Regression	C=1, penalty=l2, solver=lbfgs	0.8948
K-Nearest Neighbors	n_neighbors=15, weights=uniform, p=2 (Euclidean)	0.9009
Support Vector Machine	C=0.1, kernel=linear, gamma=scale	0.8938
Random Forest (Ensemble)	n_estimators=200, max_depth=10, min_samples_leaf=2, split=2	0.8825

4. Experiments and Results

Before modeling, EDA revealed several patterns that informed our expectations. Figure 2 summarizes the correlation structure: chest-pain type is strongly associated with disease, patients with disease have noticeably lower thalch and higher oldpeak, and the four hospitals differ markedly in disease prevalence - a fact that becomes relevant in the fairness analysis.

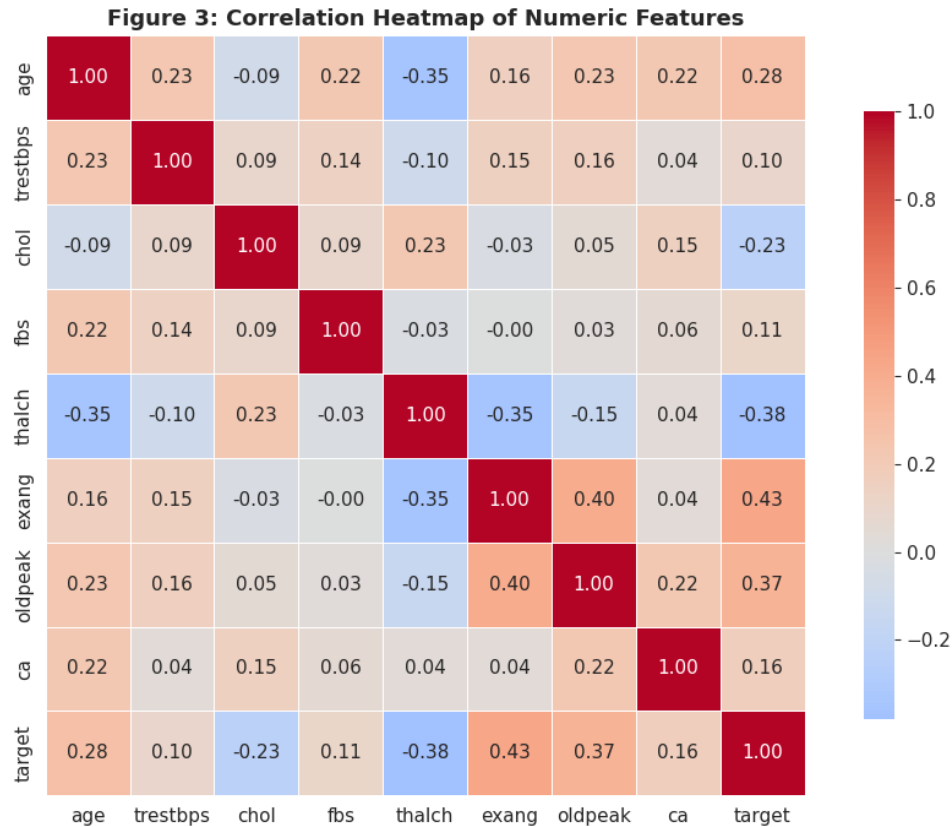


Figure 2. Correlation heatmap of numeric features with the binary target. exang (0.43), thalch (0.38), and oldpeak (0.37) are the strongest individual predictors.

Table 2. Final test-set performance for all four tuned models ($n = 184$).

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8424	0.8288	0.9020	0.8638	0.9298
K-Nearest Neighbors	0.8750	0.8624	0.9216	0.8910	0.9258
Support Vector Machine	0.8533	0.8261	0.9314	0.8756	0.9328
Random Forest (Ensemble)	0.8533	0.8378	0.9118	0.8732	0.9199

On the held-out test set, all four classifiers achieved ROC-AUC between 0.92 and 0.93, indicating that the underlying signal is strong and consistent across algorithm families. The K-Nearest Neighbors classifier was the best overall model by both accuracy (87.5%) and F1-score (0.891). SVM achieved the highest recall (0.931), which is clinically valuable because false negatives - missing a real disease case - are far more costly than false alarms. Figure 3 plots ROC curves for all four models; the curves overlap tightly, confirming that the choice of classifier matters less than the quality of the preprocessing and feature engineering.

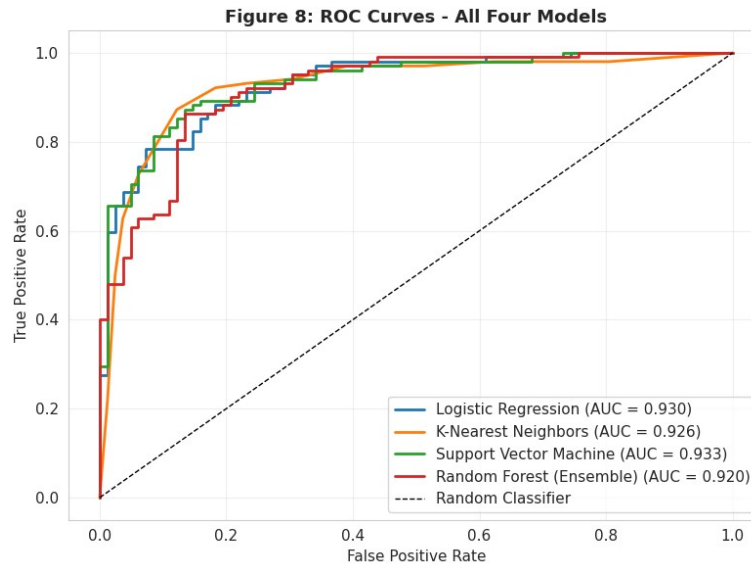


Figure 3. ROC curves on the test set. All four models clearly outperform the random baseline (dashed line), with AUC values clustered between 0.92 and 0.93.

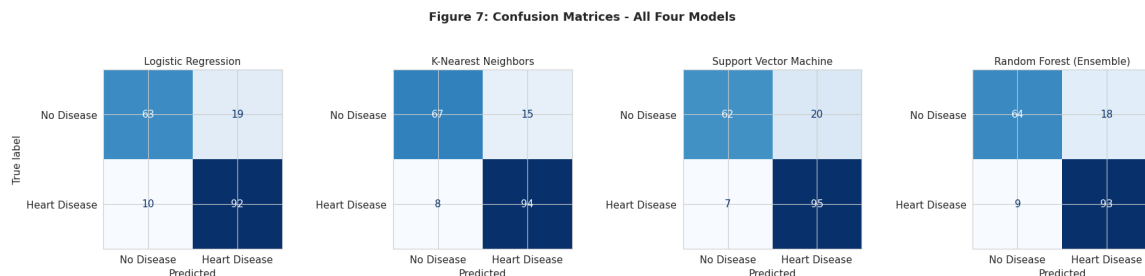


Figure 4. Confusion matrices for all four models on the test set. False negatives (bottom-left of each matrix - actually diseased patients predicted as healthy) are clinically critical because they represent missed disease cases.

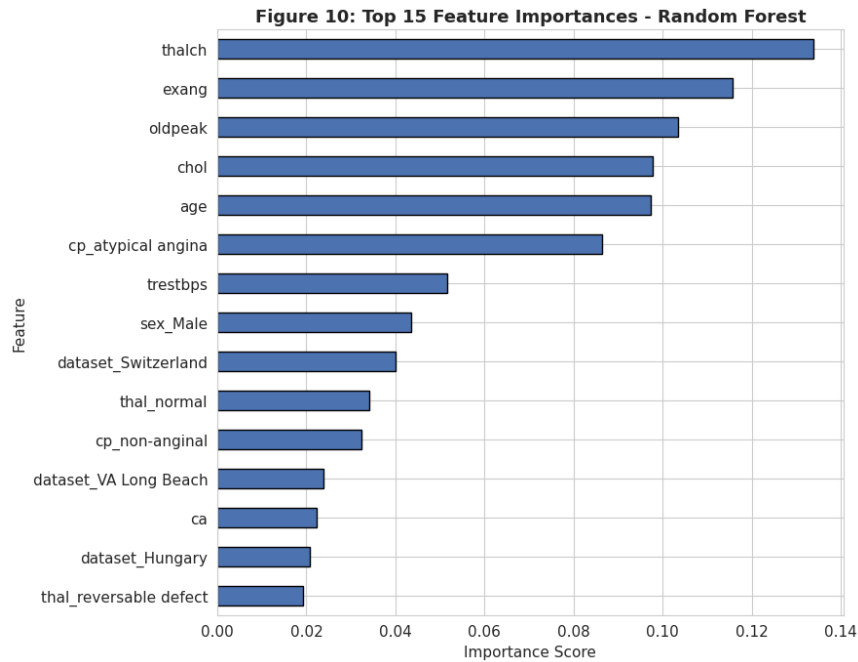


Figure 5. Top 15 feature importances from the Random Forest. The maximum heart rate (*thalch*), exercise-induced angina (*exang*), and ST depression (*oldpeak*) are the three most informative variables.

5. Discussion

An unexpected finding is that KNN outperformed the Random Forest ensemble on this dataset, despite ensembles usually dominating in classification benchmarks. Two factors likely explain this. First, after standardization the patients form fairly compact clusters in feature space, which is the ideal setting for a distance-based classifier. Second, the dataset (~918 rows) is relatively small for a Random Forest with 200 trees - the forest probably overfits the training folds slightly, as evidenced by its lower CV ROC-AUC (0.8825). Logistic Regression performed almost as well as KNN with a much simpler and more interpretable model, which would matter if we were deploying this in a clinical setting where physicians need to understand individual predictions.

Limitations include: (i) the small sample size, particularly for the Swiss and female cohorts; (ii) heavy missingness in *ca* and *thal* that we filled with imputation rather than true values; (iii) the dataset is decades old and may not reflect modern clinical practice; and (iv) we used a single random seed - variance across splits is not characterized here.

6. Data Ethics and Fairness

Medical AI raises distinct ethical concerns because errors directly affect patient health. We audited the best model (KNN) along two dimensions: patient sex and source hospital. The results are shown in Table 3 and reveal genuine fairness issues that must be disclosed before any clinical use.

Table 3. Fairness audit of the KNN model on the test set.

Subgroup	n	Accuracy	Recall
Female	35	0.943	0.875
Male	149	0.859	0.926
Cleveland	55	0.873	0.923
Hungary	65	0.846	0.739
Switzerland	26	1.000	1.000
VA Long Beach	38	0.842	1.000

Two patterns stand out. (1) Recall for female patients is 5 percentage points lower than for male patients (0.875 vs. 0.926), so the model misses more real disease cases in women. Because women constitute only 21% of the training

data, the classifier has effectively been calibrated to male physiology. (2) Recall on the Hungarian cohort drops to 0.739, far below every other hospital. Hungary is also the cohort with extensive missingness in *ca* and *thal* - two of the strongest predictors per Figure 5 - which our imputation could only partially compensate for.

Mitigation strategies for any future deployment include: collecting more female and non-Cleveland patient data; using class-weighted or fairness-aware learning algorithms (e.g., reweighting, adversarial debiasing); training hospital-specific models when sufficient local data exists; and never deploying the model as an autonomous decision maker - it must remain a screening aid that surfaces high-risk cases for clinician review. These choices are not optional refinements; they are prerequisites for ethically deploying a medical classifier in any real healthcare setting.

7. Conclusion and Future Work

We built and evaluated a complete supervised-learning pipeline for heart-disease classification on the four-hospital UCI dataset. The K-Nearest Neighbors classifier achieved the best balance of metrics (Accuracy = 0.875, F1 = 0.891, ROC-AUC = 0.926), narrowly beating SVM, Logistic Regression, and our Random Forest ensemble. Feature importance identified maximum heart rate, exercise-induced angina, and ST depression as the strongest predictors - fully consistent with the cardiology literature. The fairness audit, however, exposed performance gaps by sex and by source hospital that would block any responsible deployment.

Future work should: (i) try gradient-boosting methods (XGBoost, LightGBM, CatBoost) and stacking ensembles for further accuracy gains; (ii) use SHAP values to explain individual predictions, which is essential for clinical adoption; (iii) collect or augment the female patient subset and explore fairness-aware training; (iv) replace simple imputation with model-based imputation (e.g., MICE or k-NN imputation) for the heavily missing *ca* and *thal* features; and (v) validate the final model on truly external clinical data, not merely a held-out fold of the same dataset.

8. Team Contributions

All four team members - Abdullah Ahmed Alamoudi (2236724), Nawaf Abdulaziz Malluh (2237056), Abdullah Anwar Showail (2236114), and Ahmad Salahuddin Qadah (2340331) - contributed jointly to every stage of the project. The team worked collaboratively on dataset selection, data preprocessing and exploratory analysis, implementation and hyperparameter tuning of all four models, evaluation and fairness analysis, report writing, and preparation of the presentation slides. Code review, result interpretation, and proofreading were performed as a group to ensure each member fully understood every component of the pipeline.

9. References

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